

The `lutorch_ex` library

lookup-table FFNs: tables, projections, cartridges, and results

(draft)

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Abstract

`lutorch_ex` is a clean, cartridge-based re-implementation of the lookup-table feed-forward layer, with zero imports from the old `spiky.lutorch`. This report builds up the design top-to-bottom: the basic idea (many small discrete tables addressed by a locality-sensitive hash), the projection wrapper that makes it a drop-in FFN, the cartridge abstraction, each concrete cartridge’s math and usage, and finally a seven-way champion-topology sweep at 48k steps.

1 The basic idea: lookup tables over an LSH

A lookup-table (LUT) feed-forward layer replaces the dense MLP with many small *discrete* tables, each addressed by a locality-sensitive hash (LSH) of the input. The hash is cheap — a handful of sign bits — so a table “read” is a gather plus a few adds rather than a matrix multiply.

One table. Work in a per-head space $z \in \mathbb{R}^{d_{\text{in}}}$. A table picks `nap` *anchors* and forms `nap` scalar differences u_j ; the *signs* of the u_j form an `nap`-bit address $c \in [0, K)$ into $K = 2^{\text{nap}}$ cells, and each cell holds a learnable vector $W[c] \in \mathbb{R}^{d_{\text{out}}}$. The address is the LSH: inputs that fall on the same side of every anchor hyperplane map to the same cell. The margins u_j (how far each difference is from flipping its sign) measure how confident that address is — small margins mean a bit is near its boundary.

Anchor-pair vs single-anchor tables. Two addressing modes:

- **pairs** (`anchor_mode="pairs"`): $u_j = z[a_j] - z[b_j]$, a difference of two coordinates — a random-hyperplane LSH on coordinate pairs.
- **single** (`anchor_mode="single"`): $u_j = z[a_j]$, the raw coordinate, so the bit is just its sign — an axis-aligned LSH.

Both share the same read; only how the `nap` differences are formed changes.

Many tables, many heads. The layer stacks `tph` tables per head over G head-groups ($G = \text{n_groups} = \max(h_{\text{in}}, h_{\text{out}})$). A head’s read *sums* the addressed cell(s) over its `tph` tables (optionally weighted — see the cartridges), giving a per-group output $[B, G, d_{\text{out}}]$, which a routing step maps to $[B, h_{\text{out}}, d_{\text{out}}]$ (identity when $h_{\text{out}} = G$; a group-sum when $h_{\text{out}} = 1$).

Geometry and the base. The immutable geometry is LUTSpec: `h_in`, `h_out`, `tph`, `nap`, `d_in`, `d_out`, with `n_cells = 2nap`, `in_features = hindin`, `out_features = houtdout`. The abstract base `ManifestoLUT` owns the learnable cell tables $W \in \mathbb{R}^{G \times \text{tph} \times K \times d_{\text{out}}}$ and the addressing (`_addresses`, which uses only spec/anchors/powers — never the weight *values*), the routing invariant (`_route`), and the cell total-variation penalty (§5.2). The faithful default init draws cell tables per head from a CPU generator seeded (`seed + h + 1`), $\text{Uniform}[-\sigma, \sigma]$ with $\sigma = \text{weight_init_std}$ (realised std $\sigma/\sqrt{3} \approx 5.8\text{e-}4$ at the 10^{-3} default), reproducing the OLD LightMHL draw bit-for-bit.

2 ProjectionLUT: input and output projections

On its own a table reads a narrow per-head vector. `ProjectionMHL` (the “ProjectionLUT” wrapper) turns a cartridge into a plain $d_{\text{model}} \rightarrow d_{\text{model}}$ FFN with a linear map on each side, for two reasons:

- **input projection (compress)** — a Linear $d_{\text{model}} \rightarrow \text{in_features}$ learns the coordinates the hash reads, i.e. it *shapes the hashing* so the sign bits fall where they carry signal, instead of hashing raw model dimensions.
- **output projection (decompress)** — a Linear $\text{out_features} \rightarrow d_{\text{model}}$ lets the tables live in a narrow d_{out} space and be *compressed*: the cells are small, and the projection expands the summed read back to model width.

The forward is $z = \text{compress}(x)$ (reshaped to $[B, h_{\text{in}}, d_{\text{in}}]$); $y = \text{cartridge}(z)$; $\text{out} = \text{decompress}(y)$. Either side may be switched off (Identity) when the widths already match.

Init and first-backward warmup. `compress` $\sim \mathcal{N}(0, 0.02)$; `decompress` is *zeroed*. Since $\partial \text{out} / \partial y = W_{\text{dec}} = 0$ at init, the first backward sends no main-loss gradient to the cartridge or `compress`; only `decompress` (and the tables, via the TV term) move on step 0, and the input side “warms up” once `decompress` leaves zero — reproducing OLD’s zero-contribution FFN at init. Both inits are meta-guarded so the deployment skeleton allocates nothing.

Two capabilities it mediates. (i) `SupportsDecompressBake`: a cartridge may expose `decompress_scale()` (length `out_features`); `ProjectionMHL` folds it into the `decompress` matrix on every forward (scaling column $k = \text{scaling cartridge output channel } k$, no extra op) — the route by which a quantised cartridge’s per-channel scale moves into the dense projection. (ii) `SupportsDeploymentExport`: `to_deployment()` for compaction, used by the deployment path (§5.1); `ProjectionMHL` is the seam where the power-of-two scale is folded into `decompress`, once, at export.

3 Cartridges: different math, one contract

Everything above is fixed: the geometry, the addressing, the per-head sum, the routing, the projection wrapper. A *cartridge* is the swappable lookup *math* that sits inside that contract — how a table turns its addressed cell(s) and margins into an output. All cartridges share the read skeleton (each group addresses one cell c_g per table from the sign pattern, the read sums over the `tph` tables, then `_routes` $[B, G, d_{\text{out}}] \rightarrow [B, h_{\text{out}}, d_{\text{out}}]$); they differ in whether the read is hard or blended, whether it is weighted by a learned confidence score, and whether it is quantised. The next section covers each. **[TBD — cross-check the finer per-cartridge forms against `doc/lutorch/lut_mechanisms.tex` (§LightMHL, §confidence forms, §Gen-2 soft-sign).]**

4 The cartridges

4.1 ConfidenceLUT

Gen-3 scored read. A learned-margin confidence score weights each table’s cell contribution. With margins u and $m = |u|$, the per-table score is

$$s = \left(\sum_j m_j \right) \exp\left(\gamma \sum_j \log \sigma(\beta m_j) \right),$$

$\beta = e^{\log \beta}$, $\gamma = e^{\log \gamma}$ learnable (per layer; g frozen at 0 in the champion). A large score means every margin is large, i.e. the address is confident. The $n=2$ read blends the addressed cell c_1 and the runner-up c_2 (flip the least-certain bit, the smallest margin $u^* = \min_j u_j$) with a softmax over the deciding margin: $v = \sigma(2u^*/\tau)$, contributing $s[vW[c_1] + (1-v)W[c_2]]$; $n=1$ reads only the scored cell $sW[c_1]$. Gradients flow to W , β , γ (and τ for $n=2$).

```
from spiky.lutorch_ex import LUTSpec, ConfidenceLUT, ProjectionMHL
spec = LUTSpec(h_in=8, h_out=8, tph=64, nap=8, d_in=48, d_out=48)
cart = ConfidenceLUT(spec, seed=1, read_top_n=2, beta_init=2.0, gamma_init=1.0,
                    read_tau_init=0.5, read_tau_learnable=True)
ffn = ProjectionMHL(cart, d_model=384) # drop-in d_model -> d_model FFN
```

4.2 QuantisedConfidenceLUT (extended)

The `p2_int8` quant-aware twin of `ConfidenceLUT`. Same confidence score/blend, but the read is quantisation-aware with a straight-through estimator (STE):

- **Cell tables** are fake-quantised to int8 with a per-(group,channel) *power-of-two* scale: for group g , channel k , an integer exponent $e_{g,k}$ (`head_chan_exponents`) gives $\hat{W} = \text{round}(W/2^e) \cdot 2^e$, clamped to int8; STE passes the identity gradient to the fp32 master W .
- **Blend weights** are rounded to powers of two: $2^{k'}$ and $2^{k'-q}$ with $k' = \text{round}(\log_2 s - c_q)$ and $q = \text{round}(2|u^*|/(\tau \ln 2))$ (stop-grad exponents; the STE carries the exact $(s \cdot v, s \cdot (1-v))/s$ gradient).
- **Integer read** (`forward_int` / the deployed path): int8 cell rows, int32 shift-add in fixed point (shift = 6, `FIXED_POINT_SHIFT`), accumulator in units 2^{-6} , then scaled by 2^{e-6} per (group,channel). Equals the fake-quant forward value to round-off.

Deployment bake. `to_deployment()` emits the int8 packed table and the per-out-feature scale 2^{e-6} ; `export_deployment` folds that pure-power-of-two multiplier *exactly* into a copy of the `decompress` weight (scaling column k by 2^{e-6} only touches the fp32 exponent field). The deployed cartridge returns the *raw* int32 accumulator and reports `decompress_scale()`=None (the scale is static in the stored weight). Requires `read_top_n=2` and $h_{\text{out}} = G$ (the champion); fan-in would need a pre-route scale.

Exact constants (from `sources/quantisation_simple_norowops.md`). `quant_cfg`: $b = 8$ bits, offset $o = 0$. Blend: $q = \text{clamp}(\lfloor 2u^*/(\tau \ln 2) + \frac{1}{2} \rfloor, 0, J)$ with $J = 64$, $c_q = \log_2(1 + 2^{-q})$ for $q < C = 8$ (else 0); the second cell is dropped when $q > Q = 3$. The weight exponent $k' = \text{round}(\log_2 s - c_q)$ lives in an $L = 8$ window $[-3, 4]$ ($k_{\text{max}} = 4$, 3 bits): $k' < -3$ skips the table (weight 0, gradient kept), $k' > 4$ clamps (fires on ~ 0.001 – 0.007% of tables). The per-(head,channel) weight exponent is $e_{h,c} = \lceil \log_2 \max_{t \in h, \text{cells}} |W_t[\cdot]_c| \rceil - (b - 1) + o$ — the max over the *head’s* tables, so one exponent is shared across them (int8 $o = 0$, never saturates; int4 $b = 4, o = -1$).

FIXED_POINT_SHIFT= 6 because $k' + 6 \geq 0$ and $k' + 6 - q \geq 0$ for all $k' \geq -3$, $q \leq 3$, so the integer read is pure left-shifts-and-adds. The decompress fold is exact precisely because $e_{h,c}$ is shared across a head’s tables (it factors out of the table sum) *and* nothing sits between the table sum and decompress. Measured cost: $\sim +0.003$ bpb vs the unquantised soft read; footprint 151.0 MB fp32 $\rightarrow$ 37.75 MB int8 (4 $\times$) $\rightarrow$ 18.87 MB int4 (8 $\times$).

```
from spiky.lutorch_ex import QuantisedConfidenceLUT, ProjectionMHL, export_deployment,
    load_deployment
cart = QuantisedConfidenceLUT(spec, seed=1, quant_mode="p2_int8", read_top_n=2,
                             beta_init=2.0, gamma_init=1.0, read_tau_init=0.5)
ffn = ProjectionMHL(cart, d_model=384)
# ... train ...
export_deployment(model, "m.lxq") # int8 bake + fold pow2 scale into decompress
dep = load_deployment("m.lxq", make_skeleton, device="cuda").eval() # fp32 table never
    allocated
```

4.3 FusedManifestoHardLUT

Hard value read $\sum_t W[c_t]$ (the argmax cell per table, no score weighting), fused. The tier-2 native path uses the lprojection non-smooth backward; tier-1 fuses the per-table gather + sum with embedding_bag. [TBD — exact forward/backward from manifesto_hard.py + _native_ops.NativeHard.]

```
from spiky.lutorch_ex import FusedManifestoHardLUT, ProjectionMHL
ffn = ProjectionMHL(FusedManifestoHardLUT(spec, seed=1), d_model=384)
```

4.4 FusedManifestoSoftLUT

Soft temperature-blended read; the native path uses lprojection_forward_smooth and its native_smooth backward. [TBD — soft-blend math from manifesto_soft.py.]

4.5 FusedSoftSignHardLUT / FusedSoftSignSmoothLUT

Gen-2 soft-sign cartridges. The two-cell blend weight is $w = \sigma(-2\rho/T_{\text{sel}})$ with $\rho = |m|/(T_{\text{soft}} + |m|)$ (deciding margin m , learned temperatures $T_{\text{soft}}, T_{\text{sel}}$); the hard variant uses the straight-through value, the smooth one the true blend. The surrogate’s m/T gradients are a one-launch softsign_surrogate_grad kernel (tier-1: eager). [TBD — confirm against softsign*.py + _native_softsign.py; lut_mechanisms has the Gen-2 forms.]

```
from spiky.lutorch_ex import FusedSoftSignHardLUT, FusedSoftSignSmoothLUT, ProjectionMHL
ffn = ProjectionMHL(FusedSoftSignSmoothLUT(spec, seed=1), d_model=384)
```

5 Experiments

5.1 Architecture and implementation

The experiments drop a ProjectionLUT FFN into every block of a small GPT (6 blocks, model width $d_{\text{model}} = 384$, ≈ 68.2 M params). Two implementation notes. *Native-op resolution*: the fused cartridges reach a tier-2 CUDA fast path through one choke point, `_native_ops.native_manager()`, which resolves (1) the co-located JIT extension `lutorch_ex_lprojection` (vendored lprojection / anchor-pairs kernels, built from `cartridges/csrc/` via `cpp_extension.load`), then (2) the

pure/tier-1 torch path; the int8 read is a second JIT extension, `lutorch_ex_pow2_int8_read`. *Deployment*: `export_deployment/load_deployment` write and load a compact, cartridge-agnostic .lxq safetensors checkpoint — `make_skeleton` builds on the `meta` device (no fp32 allocation) and each compacted FFN is rebuilt from a stored format tag via the leaf `deploy_registry`, so the fp32 LUT table is never allocated and the quantised footprint drops $\sim 4\times$.

5.2 Regularisation and training recipe

Two regularisers are part of the champion recipe (applied on `lutorch_ex`, not the skeleton).

Cell total-variation ($\mathcal{R}_{TV}$, $\lambda = 10$). View each table’s $K = 2^{\text{nap}}$ cells as the vertices of an nap -dimensional hypercube ($[G\text{-tph}, \underbrace{2, \dots, 2}_{\text{nap}}, d_{\text{out}}]$). The penalty sums the squared difference (channels summed in) over every Hamming-1 cell pair — a size-2 `diff` along each of the `nap` bit-axes, each pair counted once — normalised by the pair count:

$$\mathcal{R}_{TV} = \frac{1}{n_{\text{tables}} \text{nap} 2^{\text{nap}-1}} \sum_{(c, c') \text{ Hamming-1}} \|W_c - W_{c'}\|_2^2.$$

Added to the loss as $\lambda \mathcal{R}_{TV}$ ($\lambda = 10$) through a separate backward; differentiable w.r.t. W only; adds nothing to the forward. Hamming-1 neighbours are addressed by inputs differing in a single anchor-sign bit, so penalising their squared difference smooths the table — a one-bit address flip changes the read little (a locality prior over the discrete lookup). (Bit-for-bit the reference `LightMultiHeadLUT.cell_tv`.)

Head (table) dropout ($r = 0.2$). Inverted dropout at whole-table granularity: one independent Bernoulli per (sample, group=head, table) on $[B, G, \text{tph}]$, keep prob $1 - r$, survivors rescaled $1/(1 - r)$; dropped tables contribute 0. Train-time + grad-only (eval reads all tables); the same keep-mask multiplies the backward. It discourages co-adaptation among a head’s `tph` tables.

Hyperparameters. All seven runs share the champion topology ($h=8$, `tph`= 64, `nap`= 8, $d_{\text{in}} = d_{\text{out}} = 48$, $d_{\text{model}} = 384$), seed 1, faithful small-std init, cell-TV $\lambda = 10$, head-dropout 0.2. Training: AdamW lr 3×10^{-4} , weight decay 0.1 (none on the LUT tables), 48,000 steps, cosine schedule (10% warmup, $0.1 \times \text{floor}$), sequence length 512, device batch 12, total batch 24,576 tokens, eval every 500. wandb project `Spiky`, group `lutorch_ex`; per-run artifacts (config, metrics, summary) are under `experiments/lutorch_ex/`.

5.3 The seven-config sweep

All seven runs are complete. Final `val_bpb` (best→worst):

Configuration	cartridge / mode	wandb run	final val_bpb
ConfidenceLUT $n=2$	scored, float	<code>lutorch_ex_abl47_conf_1004_1148</code>	1.10606
QuantConfidence $n=2$	scored, p2_int8	<code>lutorch_ex_abl47_quant_1004_1148</code>	1.10724
ConfidenceLUT $n=1$	scored, float, $n=1$	<code>lutorch_ex_abl47_conf_n1_vm_1004_1806</code>	1.11737
FusedSoftSignSmooth	soft-sign, smooth	<code>lutorch_ex_abl47_fss_smooth_vm_1004_1806</code>	1.12941
FusedManifestoSoft	soft	<code>lutorch_ex_abl47_fms_wk_1004_1510</code>	1.13949
FusedManifestoHard	hard	<code>lutorch_ex_abl47_fmh_wk_1004_1510</code>	1.15152

FusedSoftSignHard	soft-sign, hard	lutorch_ex_abl47_fss_hard_wk_1004_1510	1.15647
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Reading the results. The scored `ConfidenceLUT` line wins: $n=2$ is best at **1.10606**, and int8 power-of-two quantisation of it costs only +0.0012 bpb (1.10724) — consistent with the reference’s $\sim +0.003$ bpb upper bound — for a $4\times$ smaller table. Dropping the blend to a single cell ($n=1$) costs +0.0113. The non-scored families trail: the soft reads (FusedSoftSignSmooth 1.12941, FusedManifestoSoft 1.13949) beat their hard counterparts (FusedManifestoHard 1.15152, FusedSoftSignHard 1.15647). So a soft/blended read and a learned confidence score each help, and they stack.

Learning curves. [TBD — insert the `val_bpb` learning-curve figure (7 curves); data is in `experiments/lutorch_ex/*/metrics.csv`.]